

Previous Year Question Paper (Year 2024)

Linear Algebra

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (Deemed to be University)
M.Tech/ Ph.D Odd Semester Internal Exam (23-10-24)

Subject : Linear Algebra and Applications
Duration : 01 Hr.

Course Code : AM 605
Max Marks : 10

1. Let $G: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear map that rotates a vector V about the Z -axis by an angle θ . Find the Kernel and Image of the above transformation. [4]
2. Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $F(x,y) = (4x+y-z, 2x+y+z, x+y-z)$. Find the matrix representation of F relative to the basis $\{(1,0,0), (0,1,0), (0,0,1)\}$. [2]
3. Find an Orthogonal matrix P whose first row is $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$. [4]

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (Deemed to be University)
M.Tech/ Ph.D Odd Semester Internal Exam (19-09-24)

Subject : Linear Algebra and Applications
Duration : 01 Hr.

Course Code : AM 605
Max Marks : 10

➤ Answer All the Questions

1. Define a Vectorspace and hence show that the set of all odd functions forms a subspace of vector space of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$. [3]
2. Define a subspace of a vector space V . Find a basis and dimension of the subspace W of $\mathbb{R}^3$, where (i) $W = \{(a,b,c) : a+b+c=0\}$ and (ii) $W = \{(a,b,c) : a=b=c\}$ [4]
3. Show that the functions $f(t) = \sin t$, $g(t) = \cos t$, $h(t) = t$ from $\mathbb{R}$ into $\mathbb{R}$ are linearly independent. [3]

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed to be University)

M.Tech. Autumn Semester End Exam (Dec. 2024)

Subject : Linear Algebra and Applications
Duration : 03 Hrs.

Course Code : AM605
Max. Marks : 50

➤ Calculators are allowed.

1. a) Define a Vector space and an Inner product space. [2]
b) Let V be the vector space of all polynomials of degree ≤ 3 . Determine whether or not the set $S = \{t^3, t^2 + t, t^3 + t + 1\}$ spans V ? [2]
c) If A is symmetric matrix and $P^{-1}AP = D$, then show that P is an orthogonal matrix. [2]
d) Find the basis and the dimension of the space W generated by the polynomials
 $v_1 = t^3 - 2t^2 + 4t + 1, v_2 = 2t^3 - 3t^2 + 9t - 1, v_3 = t^3 + 6t - 5, v_4 = 2t^3 - 5t^2 + 7t + 5$ [4]
2. a) b) Let $w = (1, 2, 3, 1)$ be a vector in $\mathbb{R}^4$. Find an orthogonal basis for $w^\perp$. [3]
b) Check whether the following map $F: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $F(x,y,z) = (|x|, y+z)$ is linear or not linear. [3]
c) Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $F(x,y) = (x+y, x-2y, 3x+y)$. Show that G is non-singular and hence find its inverse. [4]
3. Find SVD for $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$. Give the resultant reduced matrix A after reducing the dimensionality by one, by setting the smallest singular value to zero. [10]
4. a) Discuss Overfitting and Underfitting for system of equations with examples and hence give the solution for the corresponding system of equations in both the cases. [4]
b) Give the algorithm for finding the largest eigen value and the corresponding eigen vector using power method for a given matrix. Hence find the dominant eigen value and corresponding eigen vector for $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$ by using the power method with scaling. [6]
5. Compute the QR decomposition of the matrix $\begin{pmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{pmatrix}$ using Given's rotation and Householder's transformation [10]

Probability and Statistics

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (Deemed to be University)

M.Tech/ Ph.D Odd Semester Internal Exam (23-09-24)

Subject : Statistical Modelling

Course Code : AM 604

Duration : 01 Hr.

Max Marks : 10

➤ Answer All the Questions

1. A random variable X has the density function $f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$. Find $E\left(e^{\frac{2X}{3}}\right)$. [4]
2. 10% of the tools produced in a certain manufacturing process turn out to be defective. Find the probability that in a sample of 10 tools chosen at random, exactly 3 will be defective, by using (a) the Binomial distribution (b) the Poisson approximation to the Binomial distribution. [4]
3. State Baye's Theorem and Central Limit Theorem. [2]

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (Deemed to be University)

M.Tech/ Ph.D Odd Semester Internal Exam (22-11-24)

Subject : Advanced Statistical Techniques

Course Code : AM 604

Duration : 01 Hr.

Max Marks : 10

1. The mean height of 50 male students who showed above average participation in college athletics was 68.2 inches with a standard deviation of 2.5 inches, while 50 male students who showed no interest in such participation had a mean height of 67.5 inches with a standard deviation of 2.8 inches. Test the hypothesis that male students who participate in college athletics are taller than other male students. Also find P value of the test. (5%) [5]
2. Define linearity and non-linearity in regression models and explain by using examples. [3]
3. Give the ANOVA table in case of simple and multiple regressions. [2]

Data Analysis and Visualisation

11 Sept 2024

Assessment No 1.
Sub: Data Analysis and Visualization Code: AM 610
Answer All Questions, Total Marks: 10, Duration: 1 Hr

1. If mean= 70 , median=80, and mode= 90 for a data , then the distribution will have extreme values on which side (Left or right with respect to mean). Justify your answer

2. Table. 1

P	-2	-1	1	2
Q	2	1	1	2

Compute the correlation coefficient between P and Q. Comment on the relationship between P and Q

3.

Table. 2

X1	X2	X3	X4	X5	Y
-2	1	10	9	2	6
-1	1.2	237	8.5	1	3
0	0.8	-10	8.2	-1	1
3	0.9	50	9	4	10

X1...X5 are the input variables and Y be the output variable. If you want to select only two input variables to predict Y, which input variables will you choose and why?

4. The mean marks of a class which is conducted for 100 Marks in Subject A is 95 and in Subject B is 20 Marks. Give any three inferences.
5. Explain what is composite variable and latent variables

Each question carries 2 M

Assessment No 2.

16 Sept 2024

Sub: Data Analysis and Visualization Code: AM 610

Answer All Questions, Total Marks: 10, Duration: 1 Hr. Each Question Carries 2 M

(Note: Answer Precisely)

1. Explain a Feature scaling technique which is robust to outliers along with the justification. Transform the Data A=[10, 27, 3, 17, 30, 39, 45] by applying a Min-Max Scaling
2. The main motive of the DBSCAN algorithm is outlier detection (True/ False). Justify your answer?
3. What is the intuition behind Isolation Forest in differentiating normal points from outlier points
4. Explain how a individual decision trees are learnt in Isolation Forest Algorithm
5. What is the fundamental difference between LOF and DBSCAN algorithm in detecting the outliers?

Assessment No 3.

09 Oct 2024

Sub: Data Analysis and Visualization Code: AM 610

Answer All Questions, Total Marks: 10, Duration: 1 Hr.

(Note: Answer Precisely)

1. If the principal component and eigenvalues of the data set are [1, 1, 0, 2, 3], [1, 3, 5, 2, 4], [1, 0, 2, 0, 3], [0, 3, 5, 1, 2], [1, 2, 0, 1, 0] are 1, 3, 5, 4, and 2 respectively. Reduce the dimensionality of the Dataset $\begin{bmatrix} 1 & 0 & 3 & 5 & 2 \\ 3 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 \end{bmatrix}$ such that in reduced data it can capture 80% of the variation in the Data 3 M
2. What are the advantages and disadvantages of LDA over PCA? 2 M
3. Draw the Box plot for the data [1, 10, 9, 8, 7, 6, 30, 3, 25, 11, 100]. Mention if there are any outliers in the data. 3 M
4. Explain the significance of Scatter Plot 2 M

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University), GIRINAGAR, PUNE – 411 025

END SEMESTER EXAMINATION: AUTUMN 2024

Course Name: Data Analysis and Visualization

Course Code: AM 610

Duration: 03 Hrs

Max Marks: 50

Note:

1. Before attempting, please ensure that you have the correct question paper.
2. Marks are indicated against each question. Calculators are only allowed.
3. Write clearly and legibly. All diagrams to be neatly labeled wherever applicable.
4. Answer all questions.

Q1)

[3+3+2+2]

- a) Explain what Skewness and Kurtosis convey about the dataset and how it is computed.
- b) Explain a scenario where you use mean, median, and mode for the missing values.
- c) What are the estimates which are useful in extracting the dispersion in the data?
- d) What does excess kurtosis tell about the tails of the distribution.

Q2)

[5 + 5]

- a) Explain the need for Feature scaling techniques. Transform the data {100, 10, 25, 50, 19, 80, 75, 43} using the mean normalization technique.
- b) X1, X2 and X3 are the input features and the Y is the Target variable. Classify the data sample X1=8, X2=3, X3=7 to which colour it belongs using KNN algorithm with 3 nearest neighbors.

Sl No	X1	X2	X3	Y
1	5	10	8	Red
2	10	5	6	Green
3	7	7	5	Green
4	4	6	1	Red
5	2	7	3	Red
6	6	4	6	Green

Q3)

[3+2+5]

- a) (i) What is an outlier in the data? (ii) Difference between noise and outlier? (iii) What will be the source of outliers?
- b) Detect if there is any outlier in the following data {10, 100, 90, 80, 75, 50} using the z score method
- c) Explain in detail how an outlier can be detected using the DBSCAN algorithm

Q4)

[5+5]

- a) For the data given below, rank the features X_1, X_2, X_3 based on their correlation with y

X_1	X_2	X_3	y
1	2	1	5
2	4	3	7
5	3	5	10
17	1	2	30

- b) Explain the significance of plotting the Box plot and Violin Plot with diagrams? Explain the similarities or advantages of one over the other.

Q5)

- a) Explain the working of LDA algorithm along with advantages and drawbacks.

[5 +5]

- b) If $R_1=[2, 0, 5, 6, 1]$, $R_2=[1, 0, 0, 4, 2]$, $R_3=[0, 2, 5, 1, 4]$, $R_4=[1, 0, 1, 4, 5]$, $R_1 \dots R_4$ corresponds to the Rows of the Data matrix $P_1=[1, 0, 1, 1, 3]$, $P_2=[1, 3, 2, 0, 1]$, $P_3=[10, 3, 8, 1, 4]$, $P_4=[0, 1, 4, 2, 5]$, $P_5=[1, 10, 2, 4, 2]$, are the principal components corresponding to the eigen values 4, 3, 2, 0.8, and 0.2, respectively. Reduce the dimension of the dataset such that in reduced dimension the data captures 70% of variation.

-----END OF THE QUESTION PAPER-----

Optimization Techniques

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
M.Tech Assessment-I EXAMINATION: 27th Sept, 2018.

Course (Subject) : **Advanced Optimization Techniques**
Duration : **01 hr**

Course Code : **AM-603**
Max Marks : **10**

Note: Attempt all **Two** questions from the following. All questions carry equal marks.

1. a). Write the Phases of O.R. Model.
b). Write a short note on Basic variable, Non-Basic variable and BFS.
2. a). Solve the following L.P.P.

$$\text{Minimize } Z = 20x_1 + 10x_2$$

$$\begin{aligned} \text{S.to. } & x_1 + 2x_2 \leq 40 \\ & 3x_1 + x_2 \geq 30 \\ & 4x_1 + 3x_2 \geq 60 \\ & x_1, x_2 \geq 0 \end{aligned}$$

b). Minimize $Z = 2x_1 + 2x_2$

$$\begin{aligned} \text{S.to. } & 2x_1 + 4x_2 \geq 1 \\ & x_1 + 2x_2 \geq 1 \\ & 2x_1 + x_2 \geq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY, Pune
M.Tech Assessment-I: 24th October, 2024.

Course (Subject) : **Advanced Optimization Techniques**
Duration : **01 hr**

Course Code : **AM-603**
Max Marks : **10**

Note: Attempt all the **two** questions from the following. All questions carry equal marks.

1. Solve the following L.P.P. Using Two Phase Method.

$$\begin{aligned} \text{Minimize } Z &= 4x_1 + 3x_2 + 9x_3 \\ \text{S.to. } & 2x_1 + 4x_2 + 6x_3 \geq 15 \\ & 6x_1 + x_2 + 6x_3 \geq 12 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

- 4). Using the following cost matrix, determine the optimal job assignment and also find the corresponding cost of assignments.

(Jobs)	Machines					
	A	I	II	III	IV	V
A	1	2	3	4	5	8
B	10	3	3	2	8	4
C	7	5	6	2	4	4
D	3	5	8	2	4	4
E	9	10	9	6	10	10

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY, Pune
M.Tech Assessment-III: 13th November, 2024.

Course (Subject) : **Advanced Optimization Techniques**
Duration : **01 hr**

Course Code : **AM-603**
Max Marks : **10**

Note: Attempt all the **three** questions from the following.

- 1). Write a short note on Travelling Sales man Problem. 2 Marks
- 2). A small project is composed of six activities whose time durations are listed in the table as follows: 3 Marks

Activity	1-2	1-3	2-3	2-4	3-4	4-5
Duration in days	20	10	10	12	6	10

Draw the network for the above project. Also find the Critical path

- 2) Using Kuhn-Tucker conditions solve the following NLPP. 5 Marks

$$\begin{aligned} \text{Max } Z &= -2x_1 - 3x_2 + x_1^2 + 2x_2^2 \\ \text{S.to. } & x_1 + 3x_2 \leq 6 \\ & 5x_1 + 2x_2 \leq 10 \end{aligned}$$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)

Ph.D END SEMESTER EXAMINATIONS : Nov. 2024

Course (Subject) : **Advanced Optimization Techniques** Course Code : **AM-603**

Duration : **03 hrs**

Max Marks : **50**

Note: 1. Marks are indicated against each question.

2. Before attempting , please ensure that you have the correct question paper.

3. Attempt **All Five** questions. All questions carry equal marks.

4. Students may request the invigilator for required Tables (if any).

1. a). Define Operations Research, Non-Basic variable, Basic variable & Basic Feasible Solution. (3 + 7 = 10 M)

b). Solve the following L.P.P. Using Simplex method

Maximize $Z = -x_1 + 3x_2 - 2x_3$

S.to. $3x_1 - x_2 + 2x_3 \leq 7$

$-2x_1 + 4x_2 \leq 12$

$-4x_1 + 3x_2 + 8x_3 \leq 10$

$x_1, x_2, x_3 \geq 0$

2. Apply the Travelling Salesman Algorithm to solve the following problem (10 M)

		(Cities)			
(Cities)		I	II	III	IV
	I	∞	4	9	5
	II	6	∞	4	8
	III	9	4	∞	9
	IV	5	8	9	∞

3. A small project is composed of seven activities whose time estimates are listed in the table as follows: (10 M)

Activity		1-2	1-3	1-4	2-5	3-5	4-6	5-6
time estimates in weeks	t_o	5	1	2	3	1	2	1
	t_m	6	1	4	6	1	2	4
	t_p	7	7	12	15	1	8	7

a). Draw the Project network

b). Find the critical path of the project and project duration.

c). What is the probability that the project will be completed at least 4 weeks earlier than expected. duration.

4).a). Find the extremum points and their nature of the function

(5+5 = 10 M)

$$f(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6$$

b). Minimize $f(x) = x^2 - 2.6x + 2$ in the interval $[-2, 3]$ using Fibonacci method with $N = 8$.

5 a). Using Kuhn-Tucker conditions to solve the following NLPP.

(7+3=10 M)

$$\text{Max } Z = -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2$$

$$\text{S.to. } x_1 + x_2 \leq 2$$

$$2x_1 + 3x_2 \leq 12$$

b). Write a short note on Genetic Algorithm and its applications.

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Data Structures and Algorithms

Assessment No 1.

10 Sept 2024

Sub: Data Structures and Algorithms: Code: AM 608
Answer All Questions, Total Marks: 10, Duration: 1 Hr

1. If $A = \begin{bmatrix} 2 & 3 \\ 7 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ 3 & 6 \end{bmatrix}$. Write a C program to multiply $A * B$ and print the output.

2.

Name	Age	City	CGPA
Vishnu	24	Pune	9.3
Shiva	25	Delhi	8.9

Write a C Program using structures to store the details of the above two students and print it

3. Give any two methods to allocate memory dynamically along with the Syntax

4. If there is a possibility to access the address of the variable using the &, then why do we need pointers?

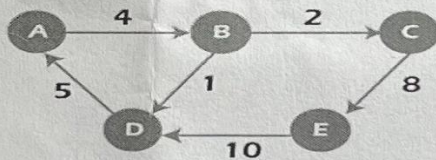
Each question carries 2.5 M each

MTech Data Science

Assessment No .2

Assessment No .2 , Subject: Data Structures and Algorithms, Answer all Questions

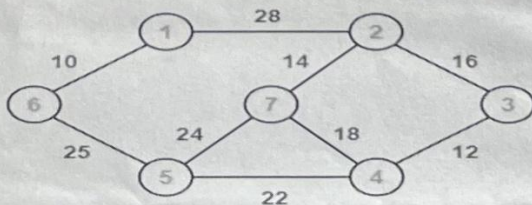
1. Explain the steps involved in inserting new data in a singly linked list considering all possible scenarios (first, last, in between). (Use either diagrams or pseudo code)
2. Mention an application that can be implemented using a. Circular linked list b. Tree c. Queue d. Hash Table
3. If the in order traversal of a binary tree gives the elements in the order: 7,1, 6,3,5, 2, 4. (assume every node has two child nodes), then draw its tree data structure, and give the post order traversal for the same.
4. Explain what is meant by AVL Tree. Mention different operations performed on the Binary search tree to make it an AVL tree with diagrams.
5. For the graph shown below, write the adjacency list and adjacency matrix representation.



Each Question carries 2M Marks Each

Assessment No 3, Sub: Data Structures and Algorithms

1. Construct the minimum spanning tree for the below graph using the Prims algorithm 3M



2. Sort the following data using the Merge Sort algorithm with clear diagrams 4 M

12	29	25	8	32	17	40
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3. The time complexity of the below program is in the order of: 3 M

```
int func(int n)
{
    if(n <= 1)
        return n;
    else
        return 3*func(n-3) - 3*func(n-2);
    end
}
```

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END SEMESTER EXAMINATION: AUTUMN 2024

Course Name: Data Structures and Algorithms

Course Code: AM 608

Duration: 03 Hrs

Max Marks: 50

Note:

1. Before attempting, please ensure that you have the correct question paper.
2. Marks are indicated against each question. Calculators are only allowed.
3. Write clearly and legibly. All diagrams to be neatly labeled wherever applicable.
4. Answer all questions.

[5+5]

- Q1) a) What are primitive data structures? Explain how it is defined in c language with examples.
b) Write a pseudo code for linear search and binary search algorithm.

[5+5]

- Q2) a) Explain the working of the Queue data structure with relevant diagrams. Write an algorithm for Enqueue and Dequeue operations.
b) Write a Pseudo code for implementing the circular linked list that can store the values a, e, and d respectively.

[5+5]

- Q3) a) Explain in detail how the data is stored in the Hash tables.
b) Explain in detail any two methods for creating Hash functions

Q 4) [5+5]

- a) Sort the following elements using the counting sort algorithm. Explain clearly with step-by-step procedure. Mention the time complexity.

4	3	5	1	1	7	2
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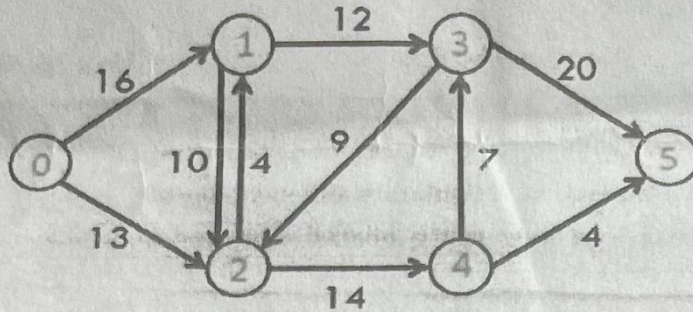
- b) Sort the following data using the quick sort algorithm. Explain clearly with step-by-step procedure. Mention the time complexity.

2	7	12	10	9	8	5
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Q 5)

[5 + 5]

- a) For the following graph find out the maximum flow: 0 be the source vertex and 5 be the Sink node



- b) What is the maximum value that can be obtained for unbreakable objects for a knapsack capable of carrying a weight of 5 kg using the dynamic programming

Item	Laptop	Guitar	Phone	Printer
weight (kg)	1	2	1	2
Value (k\$)	2	1	3	0.5

-----END OF THE QUESTION PAPER-----

Data Science: Tools and Techniques

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Assessment – I - (Autumn 2024)

Course Name: Data Science Tools & Techniques

Course Code: AM609

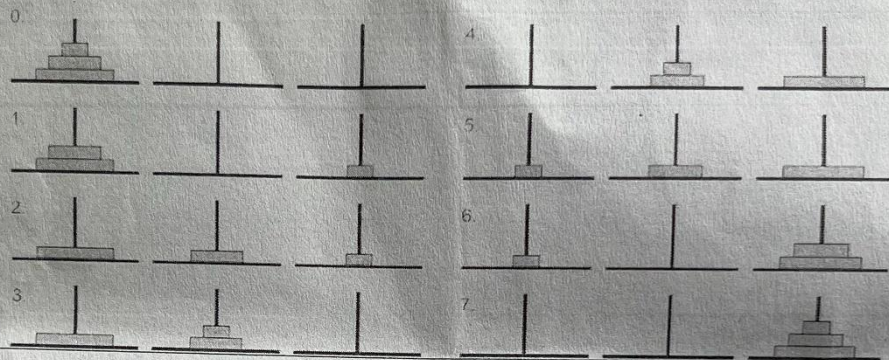
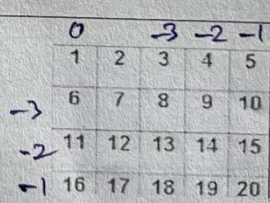
Date: 06- 09 - 2024

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly. All programs to be written in python 3.x.
2. Attempt all questions.

Q1	Write a Python program to solve the "Towers of Hanoi" puzzle for n disks. The solution should return a sequence of moves (e.g., "Move disk 1 from A to B") using only recursion. 	[03]
Q2	Write a program to input number of lines from the user and print the pattern: If the user enters 4 following patters should be printed: <pre>* ** *** ****</pre>	[03]
Q3	a) For the given matrix create a 2D list b) extract [13, 14, 15] using slicing c) extract [8,13,18] using slicing d) extract '8' value using negative indexing. 	[02]
Q4	Write a single line script using map & lambda to print a list containing the power of the said numbers: bases raised to the corresponding number in the idx lists without using pow() function. <pre>bases = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100] idx = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] Expected output: [10, 400, 27000, 2560000, 312500000, 46656000000, 8235430000000, 16777216000000000, 387420489000000000, 10000000000000000000]</pre>	[02]

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ALL THE BEST

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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
M.Tech/MS/PhD Examinations
Assessment – II - (Autumn 2024)

Course Name: Data Science Tools & Techniques

Course Code: AM609

Date: 15- 10 - 2024

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly. All programs to be written in python 3.x.
2. Attempt all questions.

Q1	a. Create a DataFrame <i>df</i> containing the random scores (between 0 and 100) of 50 students in 3 subjects: Math, Science, and English. b. Find students who scored above the mean score in at least two of the three subjects c. Display the details of these students.	[03]
Q2	Given a CSV file <i>sales_data.csv</i> with the following columns: Date, Region, Product, and Sales. Write a Python program to: a. Read the CSV into a Pandas DataFrame. b. Group the data by Region and Product and compute the total sales and average sales for each combination. c. Plot a stacked bar chart to show the total sales for each Region broken down by Product. Include appropriate labels and title.	[03]
Q3	a. Create a 3D NumPy array of shape (4, 5, 6) with random values between 100 and 200. b. Compute the mean value along axis 2 and store it in a new 2D array. c. Transpose the new array and then flatten it into a 1D array. d. Plot a scatter plot using the indices as x-values and the flattened values as y-values.	[04]

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ALL THE BEST

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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
M.Tech/MS/PhD Examinations
Assessment – III - (Autumn 2024)

Course Name: Data Science Tools & Techniques

Course Code: AM609

Date: 07– 11 - 2024

Duration: 01 Hr

Max Marks: 10

1. Marks are indicated against each question. Write clearly and legibly.
2. Attempt all questions.
3. Calculators are allowed. Exchange of calculators is not permitted.

Q1	Discuss the limitations of using Euclidean distance as a metric in high-dimensional spaces for KNN. Provide real-world examples where this metric might be misleading and suggest alternative metrics that could improve model accuracy.	[02]																																												
Q2	Consider a regression problem in which we want to predict variable y from a single feature x. We have $n>3$ data points. Consider two possible models to be estimated by linear regression. $y_i = w_0 + w_1x_i + \epsilon_i \tag{1}$ $y_i = w_0 + w_1x_i + w_2x_i^2 + \epsilon_i \tag{2}$ a) Derive the formula for estimating w_2 in the model. Assume w_0 and w_1 are known. [01] b) Will one model fit the training better than the other, will it fit equally well or is it impossible to say? Explain your reasoning. [0.5] c) Will one model fit the testing data better than the other, will they fit equally well or is it impossible to say? Explain your reasoning. [0.5] d) For the following estimator derive the update rule for w_0 by computing the gradient of the loss function. Assume learning rate to be α. [02] $w_0 + w_1x_1 + w_1x_1^3 + w_2x_2 + w_2x_2^3 + \dots + w_nx_n + w_nx_n^3$																																													
Q3	Using the dataset below, build a decision tree which classifies <i>Clicked</i> given <i>Size</i>, <i>Position</i>, and <i>Sound</i> attributes. Compute information gain and computations at each node and draw a decision tree. <table><tr><th>Clicked</th><th>Size</th><th>Position</th><th>Sound</th></tr><tr><td>F</td><td>Big</td><td>Top</td><td>No</td></tr><tr><td>F</td><td>Small</td><td>Middle</td><td>Yes</td></tr><tr><td>F</td><td>Small</td><td>Middle</td><td>Yes</td></tr><tr><td>T</td><td>Small</td><td>Bottom</td><td>No</td></tr><tr><td>T</td><td>Big</td><td>Bottom</td><td>No</td></tr><tr><td>F</td><td>Big</td><td>Top</td><td>Yes</td></tr><tr><td>T</td><td>Big</td><td>Bottom</td><td>Yes</td></tr><tr><td>T</td><td>Small</td><td>Middle</td><td>No</td></tr><tr><td>T</td><td>Small</td><td>Middle</td><td>No</td></tr><tr><td>F</td><td>Big</td><td>Top</td><td>No</td></tr></table>	Clicked	Size	Position	Sound	F	Big	Top	No	F	Small	Middle	Yes	F	Small	Middle	Yes	T	Small	Bottom	No	T	Big	Bottom	No	F	Big	Top	Yes	T	Big	Bottom	Yes	T	Small	Middle	No	T	Small	Middle	No	F	Big	Top	No	[04]
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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
Department of Applied Mathematics
M.Tech/PhD End Semester- Autumn 2024 Examinations

Roll No: - -

Date: -12-2024

Total Number of Questions: [05]

Course Title: Data Science Tools & Techniques
 Course Code: AM609

Time Allowed: 3.0 Hours

Max Marks: [50]

Instructions / NOTE

- i. Attempt All Questions.
- ii. Support your answer with neat freehand sketches/diagrams, wherever appropriate.
- iii. Assume any missing data to suit the case / derivation / answer.

Q1 [10 Marks]	Explain the following terms: (a) Confusion matrix [02] (b) Under what circumstances are Precision, or Recall better performance metrics in comparison with accuracy? [02] (c) Information Gain in Decision Trees- Classification [02] (d) K-Means Clustering [02] (e) What is the output of the following code: [02] <pre>a = 12 b = 7.4 c = 1 a -= b print(a, b) a *= 2 + c print(a) b += a * c print(b) str="Final Exam AM609" print(str[-9:-0:-1])</pre>																				
Q2 [10 Marks]	(a) Using the dataset below, build a decision tree which classifies Y given the binary variables A,B,C. Compute information gain and computations at each node and draw a decision tree. [05] <table><tr><td>A</td><td>B</td><td>C</td><td>Y</td></tr><tr><td>F</td><td>F</td><td>F</td><td>F</td></tr><tr><td>T</td><td>F</td><td>T</td><td>T</td></tr><tr><td>T</td><td>T</td><td>F</td><td>T</td></tr><tr><td>T</td><td>T</td><td>T</td><td>F</td></tr></table>	A	B	C	Y	F	F	F	F	T	F	T	T	T	T	F	T	T	T	T	F
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	(b) What is a Perceptron? Explain Perceptron Learning Algorithm?	[05]
Q3 [10 Marks]	(a) Given a CSV file sales_data.csv with the following columns: Date, Region, Product, and Sales. Write a Python program to: - Read the CSV into a Pandas DataFrame. - Group the data by Region and Product and compute the total sales and average sales for each combination. - Plot a stacked bar chart to show the total sales for each Region broken down by Product. Include appropriate labels and title. (b) Write a Python program to solve the "Towers of Hanoi" puzzle for n disks. The solution should return a sequence of moves (e.g., "Move disk 1 from A to B") using only recursion.	[05]
Q4 [10 Marks]	(a) Explain the methods get, keys, values, update, and clear in the context of dictionaries data structure. (Method explanation and syntax with example) (b) Write a program in python to sort elements of a list L in ascending order without using inbuilt functions. Eg L=[5,3,1,0,6,77] then output list should be [0,1,3,5,6,77]	[05]
Q5 [10 Marks]	(a) Explain K-Means Clustering algorithm. Discuss the limitations of using Euclidean distance as a metric in high-dimensional spaces for KNN. Provide real-world examples where this metric might be misleading, and suggest alternative metrics that could improve model accuracy. (b) What is linear regression? A linear regression requires residuals to be normally distributed. Why do we need this assumption? What will happen if this assumption does not satisfy?	[05]

Research Methodology and IPR

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
MTech EXAMINATIONS 2024

Course: Research Methodology & IPR

Code: PGC 601

Duration: 03 hr Credits: 04

Max Mark: 50

Note:

1. Marks are indicated against each question.
2. Write clearly and legibly. All diagrams (if any) to be neatly labeled.
3. Make necessary assumptions wherever required.
4. Answer all questions

Q1.a. Differentiate between research and innovation with a suitable example? Give the schematics of the research processes? **3 Marks**

b. What are the various characteristics and types of research? **3 Marks**

c. Explain the types of research ethics and what is the role of ethical committee? How TRIPS play an important role in protecting intellectual properties? **4 Marks**

Q2.a. Write notes on patentable subject matter with the conditions of patent application? What are the various steps to protect the invention? **3 Marks**

b. What do you understand from design act? How it is different from patent act? **2 Marks**

c. During the industrial interface meeting with large pool of industries, group of students mentored by a guide of DIAT developed an algorithm that shall support in differentiating the various facial gestures of human beings? Explain how the faculty coordinator will support in taking proposal to the market by protecting the intellectual properties? **5 Marks**

Q3. a. Group of students of DIAT develops a Naval material with interns and mentor from Australian University, explain the various phases of development in making this project for the global need? Explain the picture shown?

5 Marks



b. Larson, N.M., Mueller, J., Chortos, A., Davidson, Z.S., Clarke, D.R. and Lewis, J.A.*. Rotational multimaterial printing of filaments. *Nature Advance Materials*, pp.1-7, Springer NPG, vol 111, 2023? Who is the corresponding author and explain the role of the first author and how he is different from the other authors? Who is the publisher of this article and what is the name of the journal? Explain how the copyright permissions are protected? 5 Marks

Q4. a. Expand CE certification and what are the various steps to be followed for transfer of technology? Explain with a suitable example? 5 Marks

b. How publication plays an important role in the protection of intellectual properties? What are the consequences of plagiarism? 5 Marks

Q5. a. Explain phases of development during technology transfer and how intellectual properties are protected during technology transfer? What are the various methods for detecting plagiarism? 5 Marks

b. Explain WIPO and detail about the geographical indications? What are the various tips of writing a research paper? 5 Marks